Project Title: Fake Medicine Verification System

Domain: Health Tech

Team Name: MediAlert

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Department: CSE

Event: Udbhava X'26, Coding Club SR University

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Problem Scenario

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The Real-World Situation:

Counterfeit medicines are a multi-billion dollar global problem, claiming thousands of lives every year. On campus and in communities, students, faculty, and families buy medicines from pharmacies and wholesalers but have no reliable way to verify if a medicine is genuine or a "chalk pill".

The Current Failure:

Manufacturers use holographic stickers, but these are easily faked. Pharmacists and distributors often buy from unknown sources without full visibility of a medicine's journey. Consumers have no way to check authenticity. When fake medicines enter the supply chain, there is no transparent, end-to-end track & trace system, no instant verification, and no quick way to report or respond.

This leads to unsafe consumption, loss of trust, and potentially life-threatening consequences.

Our Goal

Build a **secure, transparent, and user-friendly Pharmaceutical Authenticity & Traceability System** that empowers every stakeholder to verify, trace, report, and respond—anytime, anywhere.

- **Manufacturers** — Register batches and generate unique IDs
- **Distributors/ Pharmacies** — Check-in at each supply chain point
- **Consumers** — Scan to verify authenticity instantly
- **Regulators** — Monitor, investigate, and take action
- **System** — Track, trace, report, recall, and protect lives

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Objectives & Success Criteria

Primary Objective:

To develop a secure and transparent **Pharmaceutical Authenticity & Traceability System** that prevents counterfeit medicines from reaching consumers and provides accountability across the supply chain.

Success Criteria:

- Generate unique IDs for every unit/batch of medicine.
- Provide real-time, verifiable QR code scans to check medicine authenticity.
- Detect duplicate QR codes and location anomalies to identify potential fraud.
- Enable instant reporting and recall to remove counterfeit medicines from circulation.

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PBL Statement:
Fake Medicine Verification System

Scenario

- The counterfit medicine market is a multi-billion dollar problem that kills thousands of people annually, Currently, consumers have no way to verify if the tablet they just bought at a local pharmacy is genuine or a "chalk pill." Manufacturers use holographic stickers, but these are easily faked. Pharmacists often buy from wholesalers without knowing the full pedigree of the bottle. There is no transparent, end-to-end "Track and Trace" system that allows a consumer to verify a drug's authenticity using their smartphone.

Objective

- Design a **Pharmaceutical Authenticity & Traceability System**. Focus on decentralized or secure ledger architecture and verification logic.

Functional Requirements

1. **Unique ID generation for every** medicine unit/batch.
2. **Manufacturer portal for batch registration**.
3. **Wholesaler/Distributor "Check-in" points.**
4. **Consumer "Scan to Verify"** capability.
5. Logic to detect "Duplicate Scans" (indicating a copied
6. Reporting system for "Failed Verifications".
7. Geolocation tracking of where fakes are being scanned.

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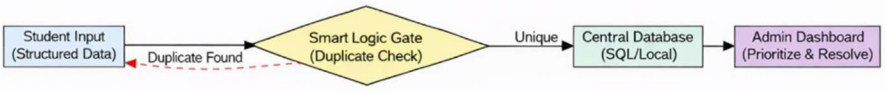
8. **Recall management system** (one-click alert to all holders of a batch).
9. **Expiry date verification logic**.
10. Role-based access for Regulators/Health Inspectors.
11. Secure API for pharmacy POS integration.

Proposed Solution Strategy



Key Solution Strategies:

1. Input Standardization: Forcing structured data entry at the source to eliminate ambiguity.
2. Role-Based Interfaces: Distinct views tailored for Students (quick reporting) and Admins (management).
3. Smart Duplicate Prevention: A logic layer that acts as a gatekeeper against redundant data.

Key Solution Strategies:

1. **Input Standardization**: Forcing structured data entry at the source to eliminate ambiguity.
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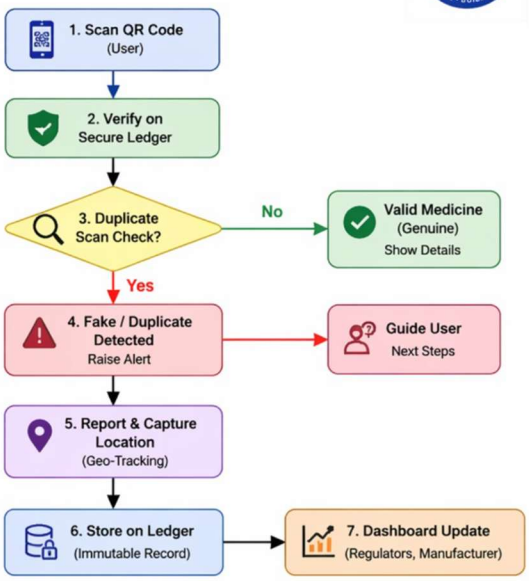
Data Flow / Workflow

1. **Submission:** User scans the QR code on the medicine package to start verification.
2. **Verification Logic:** System checks the medicine record on the secure ledger.
   Decision:
   - If valid and not scanned before → Proceed.
   - If duplicate/fake → **Stop & Alert.**
3. **Duplicate Scan Check:** System detects if the QR code has been scanned before.
4. **Alert & Reporting:** If fake/duplicate, system logs the incident, captures location, and raises an alert.
5. **Action Taken:** User and authorities are guided on the next steps.
6. **Data Storage:** All scan logs and alerts are immutably stored on the ledger.
7. **Dashboard Update:** Regulators & manufacturers see real-time updates and reports.



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PBL Statement:
Fake Medicine Verification System

Objective:

- Build a **Pharmaceutical Authenticity & Traceability** System
- Ensure **secure** and **tamper-proof** verification
- Use **QR Code / Unique ID** system
- Enable **real-time verification** using smartphones
- Focus on **security + transparency + trust**

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